

Technical Document #39: Cloud Computing - Comprehensive Overview

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Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Serverless computing allows developers to run code without managing servers.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.
- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.